

Hedge Fund Model

Mandate and investment universe

- Educational, historical AI-assisted crypto research MVP.
- Long-only, unlevered daily spot USDT universe; cash is explicit.
- Strategy sleeves: technical, econometric, ML and cross-sectional scoring.
- Future: multi-CEX automation, but no live order submission is enabled.

Hedge Fund Model

AI/ML and agent interaction

- Classical indicators are transparent baseline signals.
- Logistic regression, gradient boosting, AutoReg/GARCH and deterministic cross-sectional scoring agents are used.
- Agents emit score, confidence, horizon, cutoffs and reason codes.
- Aggregation proposes exposure; deterministic risk can veto or move to cash.

Risk Management

Taxonomy and metrics

- Market, drawdown/tail, concentration/correlation and liquidity/capacity risk.
- Model, data and operational failures are tracked separately from returns.
- Metrics include volatility, VaR/CVaR, drawdown, exposure, turnover and costs.
- Level 5 health status: ok.

Risk Management

AI support and deterministic controls

- GARCH/realized volatility and regime state inform risk budgets.
- Pre-allocation risk blocks stale, illiquid or invalid inputs.
- Post-allocation risk validates weights, turnover, concentration and cash.
- Fail-safes include volatility cash approval, invalid feature alerts and reconciliation checks.

Portfolio Management

Theory and "optimal" portfolio

- MPT and minimum variance give transparent objectives but face estimation error.
- Equal weight and inverse volatility remain robust baselines.
- CVaR/downside methods address tail risk directly.
- Submitted methods were selected on validation, not on final-test returns.

Portfolio Management

Static, dynamic and large universe

- Level 3: 5-7 assets, exactly prior 12-month estimation, static OOS hold.
- Level 4: calendar/drift/signal/risk rebalance policy.
- Level 5: point-in-time 100+ universe, top-K 25, capacity and turnover controls.
- Target weights become next-open simulated orders through one shared broker.

System Architecture

End-to-end block diagram

Frozen OHLCV -> quality gate -> causal features -> signal agents

-> orchestrator -> pre-risk -> allocator -> rebalance controller

-> post-risk -> order generator -> simulated broker -> ledger

-> metrics, monitoring, notebook and deck

Future CEX adapters are disabled; simulator artifacts are the submitted evidence.

System Architecture

Interactions, monitoring and roadmap

1. Completed daily bar creates features.
2. Agents emit typed proposals with cutoffs and confidence.
3. Risk and portfolio layers approve, cap or reject.
4. Orders fill at next open; ledger writes gross/net metrics.
5. Monitoring records freshness, drift, disagreement, incidents and runtime.

Technical Evidence

Frozen protocol and Level 5 count

- Train: 2021-2023; validation: 2024; final test: 2025.
- Accepted final lock: c33b5eb396f60b1e2a7890616b8d9ae1cd69e91375dec925b68b6673d843af5e.
- Final-test exposure: EXPOSED.
- Level 5 final count: 120 eligible, 120 scored, 25 selected.
- Runtime: 78.4s; peak RSS: 847.6 MiB.

Results and Limitations

Net final-test evidence

Level | Selected | Net return | Sharpe | MDD

Level 1 | SMA baseline | -7.4% | -0.17 | -18.5%

Level 2 | agent_ensemble | -0.6% | -0.52 | -1.4%

Level 3 | cvar_downside | -18.0% | -0.02 | -45.2%

Level 4 | calendar_monthly | -4.1% | -0.88 | -9.1%

Level 5 | large_universe_dynamic | -28.0% | -0.22 | -42.2%

Limitations: active-market survivorship/delisting bias, daily-bar liquidity proxy, short late-December 2024 Level 5 validation proof window, cash-heavy risk behavior, and Level 5 top-K benchmark rather than a full eligible-universe basket. Results did not establish robust alpha.